

BECE102L – Digital Systems Design

Module 5 –Design of Sequential Logic Circuits

Latches, Flip-Flops - SR, D, JK & T, Buffer Registers, Shift Registers - SISO, SIPO, PISO, PIPO, Design of synchronous sequential circuits: state table and state diagrams, Design of counters: Modulo-n, Johnson, Ring, Up/Down, Asynchronous counter. Modeling of sequential logic circuits using Verilog HDL..

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SEQUENTIAL CIRCUITS

- There are two types of sequential circuits:

- ❖ *synchronous*: outputs change only at specific time

- ❖ *asynchronous*: outputs change at any time

- **Multivibrator**: a class of sequential circuits.

They can be:

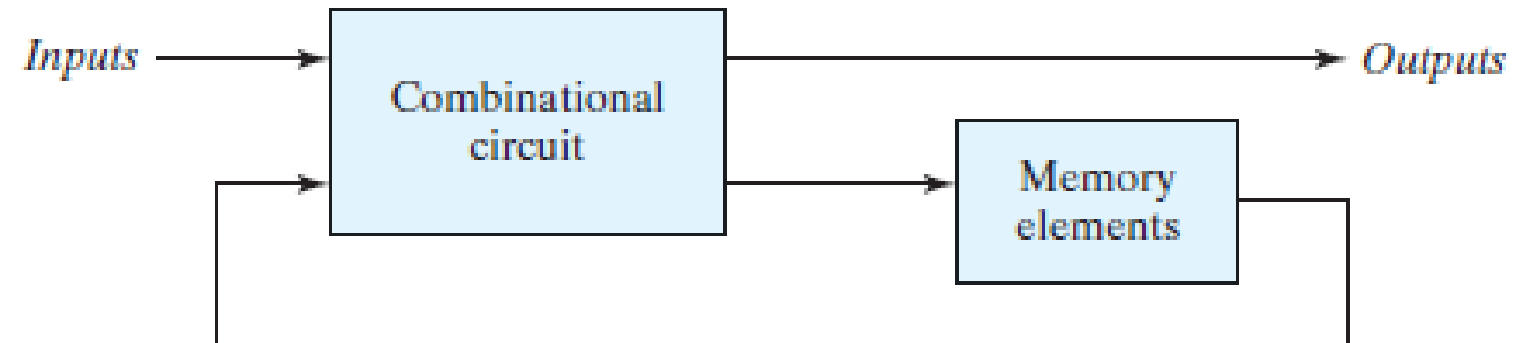
- ❖ *bistable* (2 stable states)

- ❖ *monostable* or *one-shot* (1 stable state)

- ❖ *astable* (no stable state)

- Bistable logic devices: *latches* and *flip-flops*.

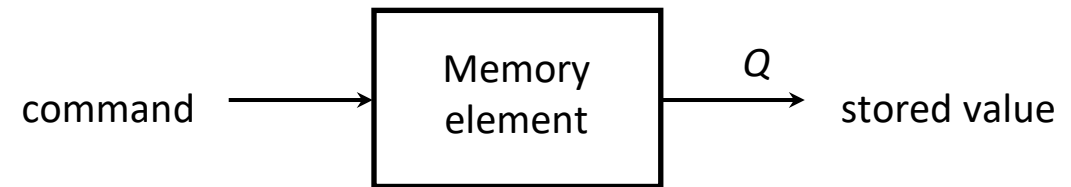
- Latches and flip-flops differ in the method used for changing their state.



Sequential circuit = Combinational logic + Memory Elements

Memory Elements

Memory element: a device which can remember value indefinitely, or change value on command from its inputs.



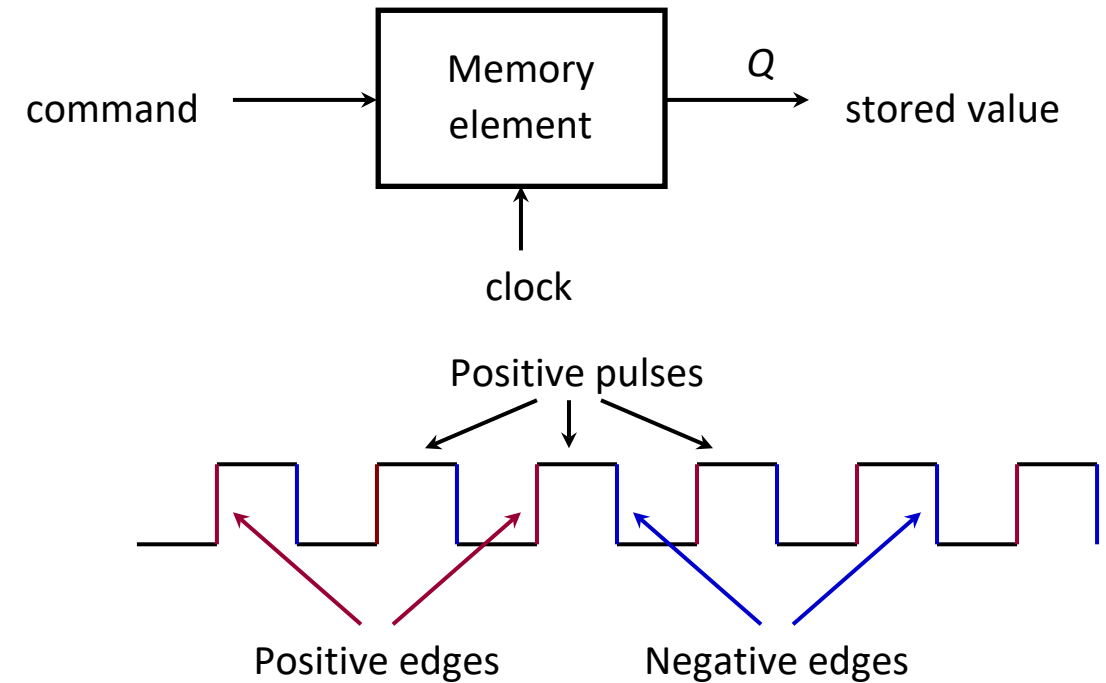
■ Characteristic table:

Command (at time t)	$Q(t)$	$Q(t+1)$
Set	X	1
Reset	X	0
Memorise / No Change	0	0
	1	1

$Q(t)$: current state
 $Q(t+1)$ or Q^+ : next state

Memory Elements with clock

- Two types of triggering/activation:
 - ❖ pulse-triggered
 - ❖ edge-triggered
- Pulse-triggered
 - ❖ latches
 - ❖ ON = 1, OFF = 0
- Edge-triggered
 - ❖ flip-flops
 - ❖ positive edge-triggered (ON = from 0 to 1; OFF = other time)
 - ❖ negative edge-triggered (ON = from 1 to 0; OFF = other time)



Latches and Flip-Flops

There are two types of memory elements based on the type of triggering that is suitable to operate it.

- Latches
- Flip-flops

Latches operate with enable signal, which is **level sensitive**. Whereas, flip-flops are **edge sensitive**.

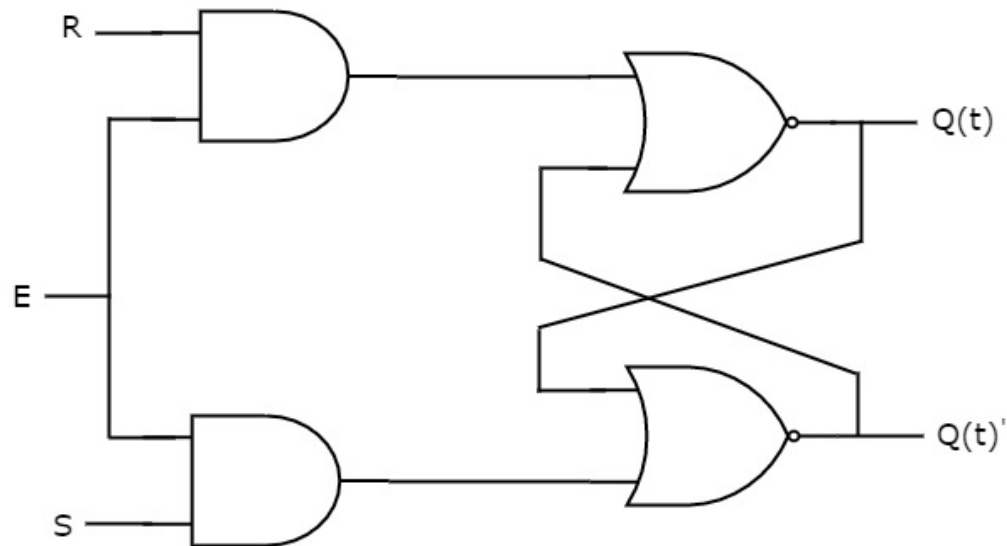
Latches and flip-flops are sequential logic circuits that can store one bit of data. They are the building blocks of more complex sequential circuits, such as registers, counters, and sequencers.

Latches	Flip flops
Latches are level triggered devices.	Flip flops are edge triggered devices.
No further classification for latches.	Flip flops can be classified as asynchronous and synchronous.
Latches can be clocked or clockless.	Flip flops are always clocked.
Latches are operated only with binary inputs.	Flip flops are worked with both clock and binary inputs.
Latches are faster as compared to flip flops.	Flip flops are slow compared to the latches.
Latches are constructed from logic gates.	Flip flops are constructed from latches along with an additional clock signal.

Types of latches

There are two main types of latches:

- **SR latch:** SR Latch is also called as Set Reset Latch. This latch affects the outputs as long as the enable, E is maintained at '1'.
- The basic SR latch has two inputs, S and R, and two outputs, Q and Q'. The output Q is the complement of Q'.

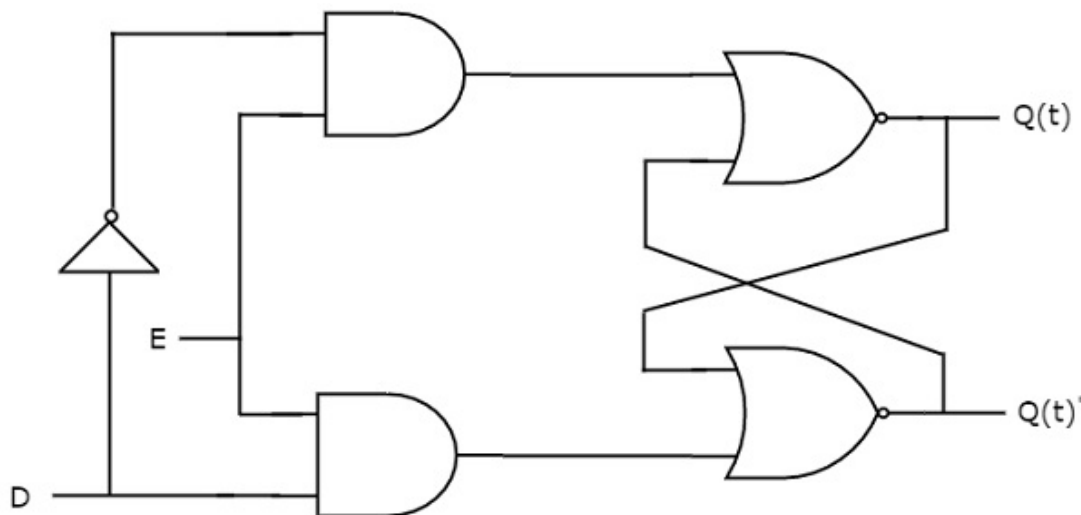


state table

S	R	Q _{t+1}
0	0	Q _t
0	1	0
1	0	1
1	1	x

•There is one drawback of SR Latch. That is the next state value can't be predicted when both the inputs S & R are one. So, we can overcome this difficulty by D Latch. It is also called as Data Latch.

•**D latch:** The D latch has one input, D, and one output, Q. The latch will set its output to $Q=D$ on the rising edge of the clock signal.



state table

D	Q_{t+1}
0	0
1	1

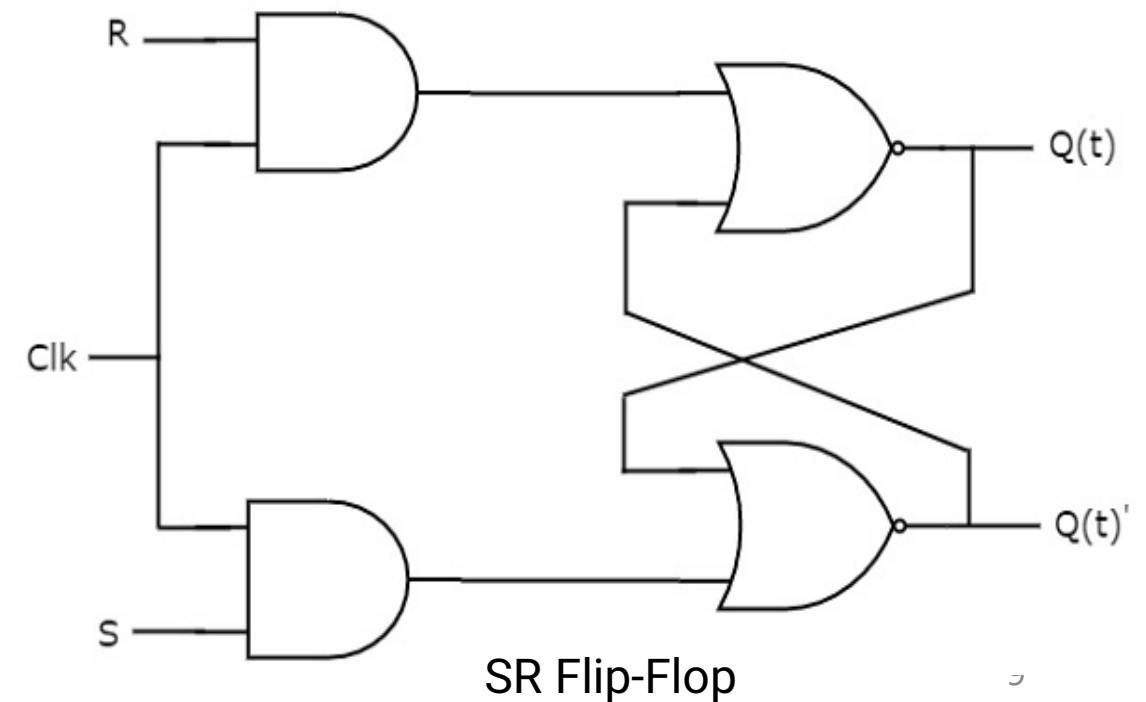
Flip-Flops

A flip-flop is a sequential logic circuit that stores one bit of data and has a clock input. The flip-flop will change its output to Q on the rising or falling edge of the clock signal, depending on the type of flip-flop.

SR flip-flop operates with only positive clock transitions or negative clock transitions. Whereas, SR latch operates with enable signal.

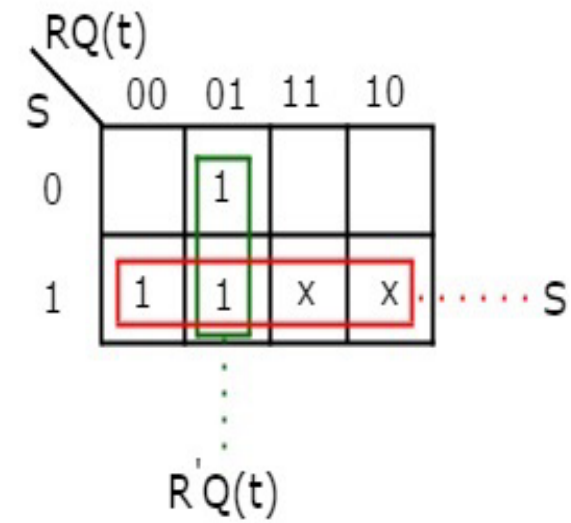
state table

S	R	Q_{t+1}
0	0	Q_t
0	1	0
1	0	1
1	1	Invalid



Characteristic Table :SR flip-flop

Present Inputs		Present State	Next State
S	R	Q_t	Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x



$$Q(t+1) = S + R'Q(t)$$

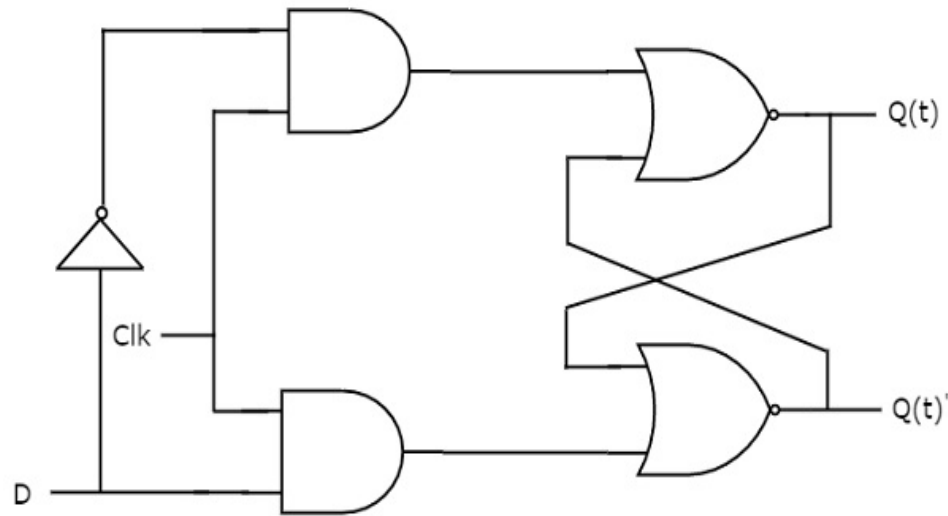
Modelling of Positive Edge Triggered SR-FF

```
module SR_FF(s,r,clk,reset,q,qb);  
    input s,r,clk,reset,  
    output q, qb  
    reg q, qb;  
  
    always @ (posedge clk)  
        begin  
            if (reset)  
                begin  
                    q <= 0;  
                    qb <= 1;  
                end  
        end
```

```
    else  
        begin  
            if (s != r)  
                begin  
                    q <= s;   qb <= r;  
                end  
            else if (s == 1 && r == 1)  
                begin  
                    q <= 1'bx; qb <= 1'bx;  
                end  
            else  
                begin  
                    q <= q;   qb <= qb;  
                end  
        end  
    end  
end  
endmodule
```

D Flip-Flop

- D flip-flop operates with only positive clock transitions or negative clock transitions.
- D latch operates with enable signal.
- The output of D flip-flop is insensitive to the changes in the input, D except for active transition of the clock signal.



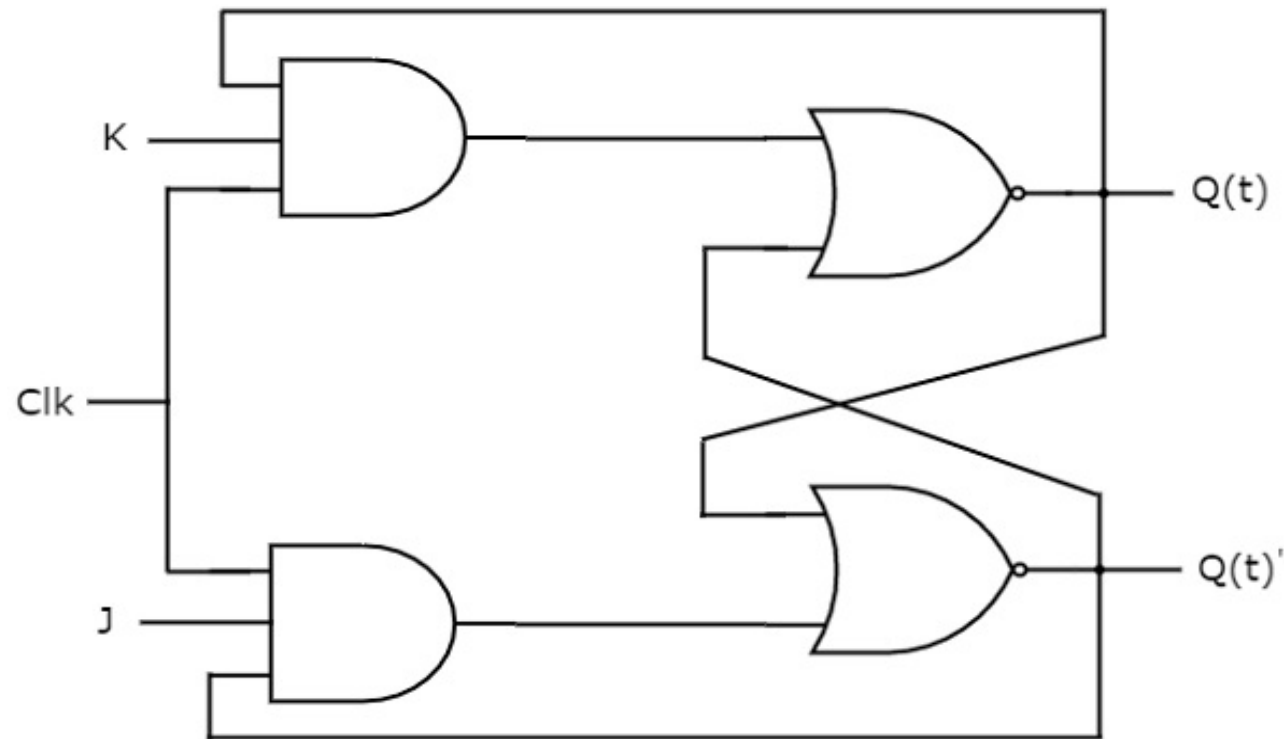
state table of D flip-flop

D	$Q_t + 1$
0	0
1	1

$$Q_{t+1} = D$$

JK Flip-Flop

- JK flip-flop is the modified version of SR flip-flop. It also operates with only positive clock transitions or negative clock transitions.

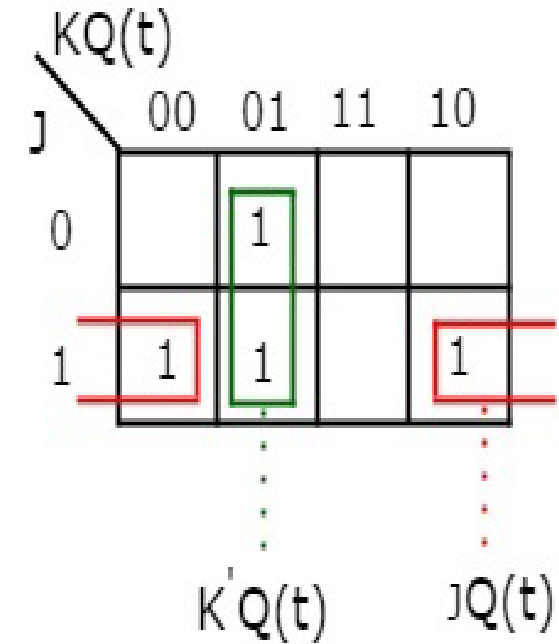


state table of JK flip-flop

J	K	Q _{t+1}
0	0	Q _t
0	1	0
1	0	1
1	1	Q _t '

Characteristic Table : JK flip-flop

Present Inputs		Present State	Next State
J	K	Q _t	Q _{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



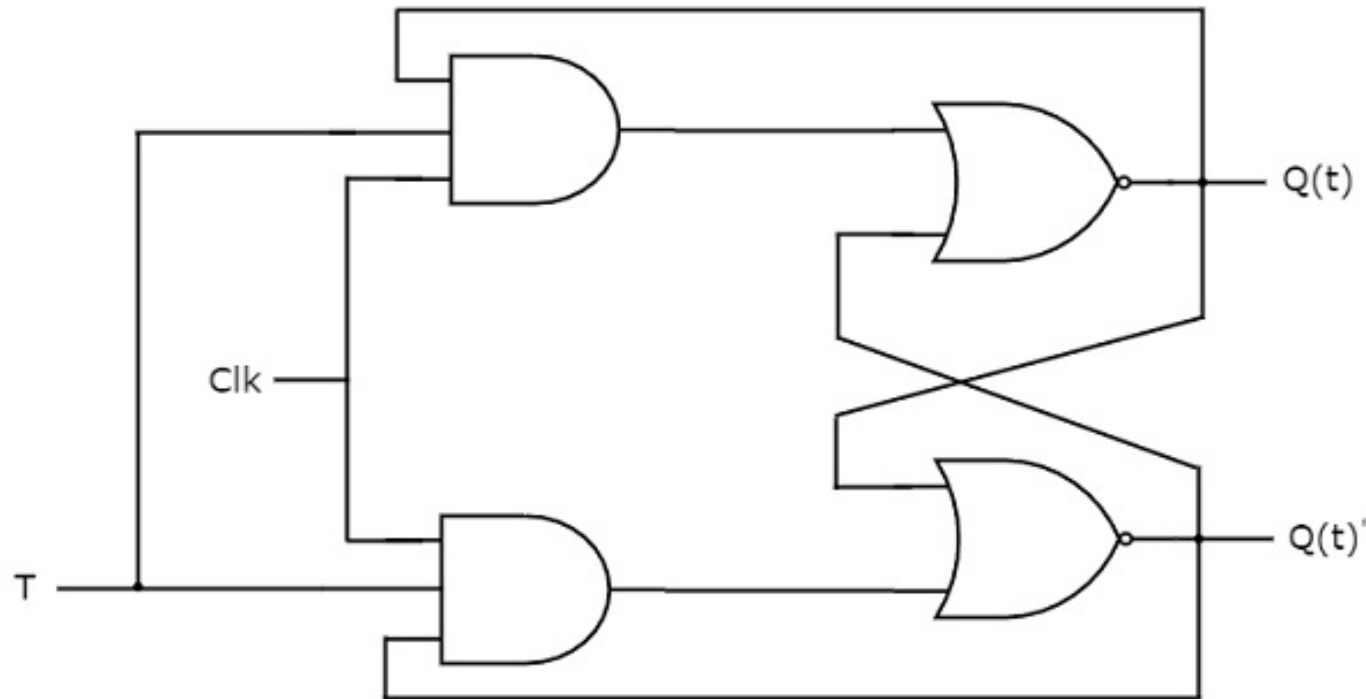
$$Q(t+1) = JQ(t)' + K'Q(t)$$

Modelling of Positive Edge Triggered JK-FF

```
module jkff_beh(clk,j,k,q,qn) ;  
input clk,j,k;  
output q,qn;  
wire q,qn;  
reg t;  
always@ (posedge clk)  
begin  
if (j==0 && k==0)  
t<=t;  
else if(j==0 && k==1)  
t<=0;  
else if(j==1 && k==0)  
t<=1;  
else  
t<= ~t;  
end  
assign q = t;  
assign qn = ~t;  
endmodule
```

T Flip-Flop

T flip-flop is the simplified version of JK flip-flop. It is obtained by connecting the same input 'T' to both inputs of JK flip-flop.



T	Q_{t+1}
0	Q_t
1	Q_t'

Characteristic Table :T flip-flop

Inputs	Present State	Next State
T	Q _t	Q _{t+1}
0	0	0
0	1	1
1	0	1
1	1	0

$$Q(t+1) = T'Q(t) + TQ(t)'$$

$$Q(t+1) = T \oplus Q(t)$$

Modelling of Positive Edge Triggered T-FF

```
module tff(t,q,clk);  
input t,clk;  
output q;  
reg q;  
always @ (posedge clk)  
if (t==0)  
q<=t;  
else  
q<=~t;  
endmodule
```